

Building Neural Network From Scratch

Part 1: Coding Neurons And Layers

Coding Our First Neuron: 3 INPUTS

```
In [3]: inputs = [1, 2, 3]
weights = [0.2, 0.8, -0.5]
bias = 2

outputs = (inputs[0]*weights[0]+inputs[1]*weights[1]+inputs[2]*weights[2] + bias
print(outputs)
```

2.3

```
In [ ]:
```

Coding Our second NEURON: 4 INPUTS

```
In [6]: inputs = [1.0, 2.0, 3.0, 2.5]
weights = [0.2, 0.8, -0.5, 1.0]
bias = 2.0
outputs = (inputs[0]*weights[0]+
            inputs[0]*weights[0]+
            inputs[0]*weights[0]+
            inputs[0]*weights[0] + bias)
print(outputs)
```

2.8

```
In [ ]:
```

Layer of Neurons

```
In [25]: inputs = [1, 2, 3, 2.5]

weights = [[0.2, 0.8, -0.5, 1],
           [0.5, -0.91, 0.26, -0.5],
           [-0.26, -0.27, 0.17, 0.87]]

weights1 = weights[0] # List of weights associated with the first neuron : w11,
weights2 = weights[1] # List of weights associated with the second neuron : w21
weights3 = weights[2] # List of weights associated with the third neuron : w31,

biases = [2, 3, 0.5]

bias1 = 2
bias2 = 3
bias3 = 0.5

outputs = [
    # Neurons 1:
```

```

inputs[0]*weights1[0]+
inputs[1]*weights1[1]+
inputs[2]*weights1[2]+
inputs[3]*weights1[3]+ bias1,

# Neurons 2:
inputs[0]*weights2[0]+
inputs[1]*weights2[1]+
inputs[2]*weights2[2]+
inputs[3]*weights2[3]+ bias2,

# Neuron 3:

inputs[0]*weights3[0]+
inputs[1]*weights3[1]+
inputs[2]*weights3[2]+
inputs[3]*weights3[3]+ bias3
]

print(outputs)

```

[4.8, 1.21, 2.385]

Using Loops for better and easier coding

```

In [50]: inputs = [1.1, 2.2, 3.3]

## List of weights
weights = [[2,0,0],
           [2,2,0],
           [2,2,2]]

# List of biases
biases = [0,0,1]

# Output of current Layer

layer_outputs = []

# For each neuron
for neuron_weights, neuron_bias in zip(weights, biases):
    # Zeroed output of given neuron
    neuron_output = 0
    # For each input and weight to the neuron
    for n_input, weight in zip(inputs, neuron_weights):
        # Multiply this inputs by associated weight
        # And add to the neuron's output variable
        neuron_output += n_input*weight # w11*x1 + w12*x2 + w13*x3 + w14*x4
    # Add bias
    neuron_output += neuron_bias ### w11*x1 + w12*x2 + w13*x3 + w14*x4 B1
    # Put neuron's results to the layers output list
    layer_outputs.append(neuron_output)
print(layer_outputs)

```

[2.2, 6.6000000000000005, 14.2]

```
In [32]: inputs = [1, 2, 3, 2.5]

# List of weights
weights = [
    [0.2, 0.8, -0.5, 1],
    [0.5, -0.91, 0.26, -0.5],
    [-0.26, -0.27, 0.17, 0.87]
]

# List of biases
biases = [2, 3, 0.5]

# Output of current layer
layer_outputs = []

# For each neuron
for neuron_weights, neuron_bias in zip(weights, biases):
    # Zeroed output of given neuron
    neuron_output = 0

    # For each input and weight to the neuron
    for n_input, weight in zip(inputs, neuron_weights):
        # Multiply input by weight and add to output
        neuron_output += n_input * weight

    # Add bias
    neuron_output += neuron_bias

    # Append neuron's output
    layer_outputs.append(neuron_output)

print(layer_outputs)
```

```
[4.8, 1.21, 2.385]
```

```
In [ ]:
```